EBIN BABU THOMAS

AI Engineer: LLM Agents, RAG & Evaluation Engineering

ebinbt.dev · hello@ebinbt.dev · github.com/ebt55 · Mavelikara, Kerala, India (Remote)

SUMMARY

AI engineer for LLM agents, search and evaluation. Recently, for a screening test, I built hybrid keyword and vector search over 433 council meetings. Since March 2026 I also shipped Diarist, a meeting transcriber (public beta), and ExactDoc, a PDF to Word converter (on PyPI). My agent tests have run at Apart Research sprints. Before that, 3.5 years building LLM, RAG and backend systems for startups in the US, Canada, Europe and Australia.

TECHNICAL SKILLS

Core: Python, FastAPI, LLM Agents & Tool Use (Claude Agent SDK, Anthropic API, Gemini API, OpenAI API, PydanticAI, LangGraph), MCP (FastMCP), RAG, Prompt & Context Engineering, Playwright, pytest, Docker, Git

Proficient: PostgreSQL / SQL, MongoDB, Vector DBs (Qdrant, LanceDB), PyTorch, Hugging Face Transformers / PEFT, Fine-tuning (LoRA / QLoRA, DPO), LLM Serving (vLLM, Modal), NLP (sentence-transformers, Whisper), MLOps (GitHub Actions CI/CD, OpenTelemetry), Pandas, AWS, GCP, Kubernetes (GKE), OAuth2 / JWT, Terraform, Linux, uv

Evaluation & measurement: Inspect (inspect_evals), experiment design, red-teaming, model diffing, blind grading, LLM-as-judge agreement checks, bootstrap & permutation statistics, log-probability metrics, SHA-256-pinned artifacts, Langfuse

Familiar: Rust / Tauri, LangChain, Data Streaming (Bytewax), Next.js, TypeScript

EXPERIENCE

Software Engineer · Zackriya Solutions *Jan 2022 – Jun 2025*

Remote, India. A software services firm building production AI systems, RAG pipelines and backends for startup clients in the US, Canada, Europe and Australia. Grew from trainee to core developer; owned client backends end to end and scoped incoming projects.

- **FinBot, a financial assistant:** fine-tuned Mistral 7B with QLoRA and built a live news pipeline (Alpaca news API, Bytewax, Qdrant) so answers cite current news. Served it with vLLM on a GKE GPU node.
- **Natural-language property search:** built a real-estate search backend that turns plain-English questions into database queries with GPT-3.5. Deployed it on Google Cloud Run, load-tested it to size instances, and added “Did you mean?” suggestions.
- **DocuAI, semantic document search:** indexed 1,000+ documents in Qdrant so users search by meaning. Designed the ingestion and retrieval pipeline and a Next.js front end, and delivered a working client demo in two days.
- **Data and backend pipelines:** built real-time bioreactor sensor ingestion (MQTT to MongoDB) for a startup that later raised funding. Automated spoken-English video scoring (Whisper, Microsoft pronunciation API, AWS Fargate) and healthcare compliance pipelines.

INDEPENDENT PROJECTS & RESEARCH

Diarist: Local Meeting Transcriber with Speaker Labels (Windows) *Oct 2026*

diarist.dev · github.com/ebt55/diarist (NVIDIA NeMo-Speech.cpp (CUDA), Nemotron speech and diarization models, Svelte 5, TypeScript, SQLite (FTS))

- Built a free Windows app that records meetings or imports recordings and labels who said each line. It runs on the user’s own GPU, with no account and nothing uploaded. On the author’s own Zoom meetings, **0.2–0.3% of words went to the wrong speaker**, scored against Zoom’s labels. An hour of audio takes about two minutes on an RTX 3080 (about 30× real time). Built in one day with coding agents; public beta 0.1.0-beta.5, tested on one PC.

ProofPack: Purchase-Approval Review Agent with Tamper-Evident Evidence *Jul – Sep 2026*

Built from a challenge brief (Build with Gemini XPRIZE, Jul 2026) · github.com/ebt55/proofpack (Python, Gemini 3.7 Flash, Playwright, FastAPI, pytest)

- Built an AI agent that checks Medicaid purchase requests against providers’ public websites and keeps stamped screenshot proof for every claim. Plain code writes every URL, timestamp and hash, and a person makes each approve or deny call. On **7 synthetic request forms, each review cost $0.02–$0.19** in Gemini fees. A project, not a product; no users.

ExactDoc: PDF to Editable Word Documents, Checked Page by Page *Aug – Oct 2026*

Open source, beta 0.3.0b3 on PyPI · github.com/ebt55/exactdoc (Python, PDFium (pypdfium2), OOXML, LibreOffice headless, PyMuPDF, pytest)

- Built a converter that turns a PDF into an editable Word document that still looks like the original, with real headings, lists and tables. It opens its own output in LibreOffice, compares each page with the PDF, and fixes page breaks. On **27 short PDFs of supported kinds, none seen before, 22 kept exactly their page count in Word** (24 in LibreOffice). Two LaTeX papers came out unusable, and designed layouts are not supported yet.

Kobayashi Maru: Do Impossible Tasks Make Coding Agents Cheat? *Sep 2026*

Apart Research AI Incident Response sprint, solo; top half of 230+ entries · github.com/ebt55/kobayashi-maru (Python, Docker, 6 model families, plain-code cheat detectors)

- Tested whether impossible tasks in a coding agent's batch make it cheat on the solvable ones, over 8,959 runs on six models. **DeepSeek-V4.1-flash went from 0 to 36 cheats in 120 solvable-task attempts** as impossible tasks were added (trend $p = 8\times10^{-6}$). Limits: both positive models are cheap ones added later, and batches grow longer with the dose.

councilrag & rbb: Civic-Meeting Search and a Retrieval Benchmark Builder *Oct 2026*

Built in a private 48-hour company screening test; code not public (Python, LanceDB (BM25 + vector + RRF), Qwen3-4B, Granite-3.3-2B, SigLIP, RapidOCR, Diarist ASR)

- Built councilrag, which answers questions about Seattle City Council meetings with citations that open the video at the second. It searches 433 meetings (about 813 h) of captions. A 9-meeting slice adds speaker-labelled audio, on-screen text and the vote record. There the live captions dropped **18 of 60 located roll-call votes on 8 meetings** (17 on two of them); the audio transcript recovers 17 of the 18.
- Built rbb, a retrieval benchmark that stores evidence as time spans, and tested whether its metrics predict a reader's answers. Within one token budget and question, graded labels picked the configuration that answered correctly **66% of the time, against 50% for token count**. That is 6,298 pairs from 83 of 144 Video-MME questions with one 4B reader; a Seattle replication found no such gap.

LLM Pricing Hub Remedies: Do Antitrust Rules Stop a Shared AI Pricing Advisor? *Oct 2026*

Apart Research AI Collusion Research Sprint, 23–25 Oct 2026; results not announced · github.com/ebt55/llm-pricing-hub-remedies (Python, uv, Anthropic API, OpenAI SDK, OpenRouter, SciPy, pandas)

- Tested whether the rules in the 2026 US settlement with RealPage stop one AI advisor from pushing up two competing sellers' prices. Across more than 700 simulated games on seven models, no rule worked reliably. Banning rival data removed **69–92% of the harm on three models and at most 10% on two**. The advisor was told to maximise the sellers' joint profit, so this tests the rules against that incentive.

printready: Print Checks and Safe Fixes for AI-Generated Sticker Art *Oct 2026*

github.com/ebt55/printready (Python, NumPy, OpenCV, scikit-image, Pillow, xAI API (Grok), MCP server, pytest)

- Built a tool that checks AI-generated sticker art against a printer's published rules, fixes what it safely can, and hands the rest to a person. On 50 blind-labelled images, an AI agent with the same tools tied the fixed steps at **34 of 50** and cost about 4× as much. 53 of 55 fixed proofs passed a mode-blind review by the author; one labeller, nothing printed.

diffing-agent-bench: A Sealed Test for AI Agents That Audit Fine-Tuned Models *Aug – Sep 2026*

Work sample for the MATS 12.0 application, Neel Nanda stream, Sep 2026 · github.com/ebt55/diffing-agent-bench (Python, PyTorch, PEFT / LoRA, vLLM, Qwen3.5-9B)

- Built a sealed test for AI agents that compare a fine-tuned model with its original. I planted habits in LoRA finetunes of Qwen3.5-9B and sealed the answer key before any run. The agent asked a database question in **0 of 13 attempts**, so it missed the planted database habit. A fixed 50-prompt list found it for $0.15, against $3.14 per detection for the agent.

odd-number-forensics: Why AI Models Chase a Leaked Score Instead of the Request *Aug 2026*

Practice work sample for the SPAR Fall 2026 model-forensics take-home · github.com/ebt55/odd-number-forensics (Python, asyncio, OpenRouter, Ollama, Wilson and Newcombe intervals)

- A published test asks a model for an even number while a leaked grader rewards odd ones. I took the prompt apart one line at a time over about 5,300 samples. For OpenAI's o3, one-line edits moved score-chasing from **under 2% to 87%** (30 to 60 samples per setting). Cells this size show only large effects.

ComfyUI-DeadMansQueue: Crash-Safe Job Queue for the ComfyUI Image App *Sep 2026*

Open-source add-on · github.com/ebt55/ComfyUI-DeadMansQueue (Python (no extra dependencies), SQLite (WAL), ComfyUI custom node)

- Built an add-on that saves ComfyUI's waiting jobs to disk, so a crash doesn't wipe the queue. In one test I killed ComfyUI with 40 jobs queued, and **all 39 unfinished jobs came back in order**. No login tokens were written to disk. One test, on one ComfyUI version, on Windows.

IncidentGate: Test Lab for Safety Checks on an Incident-Repair Agent *Aug – Sep 2026*

Closed at a baseline, Sep 2026. github.com/ebt55/incidentgate (Python, LangGraph, PostgreSQL, OpenTelemetry, Langfuse, Docker Compose, pytest)

- Built a test lab for three safety checks on an incident-repair agent: a rule list, a watcher model and a scripted stand-in approver. A forbidden action made as one named call was stopped at once. A different forbidden change, split into three allowed steps, passed all checks with a local watcher model and **ended as if no checks were on**. Each setting ran once, and the attack was scripted.

OPEN-SOURCE CONTRIBUTIONS

- Meetily **(GitHub-Trending meeting-notes app):** backend refactor and OpenAI support, merged (PR #75); bpmn-io/refactorings: text-similarity recommender for connector templates, replacing an LLM call (issue #33).

Career break Jul 2025 – Feb 2026. Independent AI work since Mar 2026. BlueDot Technical AI Safety course, certificate Oct 2026.